

Exponential (1)

Exercise 1:

Solve the following equations and inequalities:

1) $e^{3x-1} = 1$

4) $4e^{-x} + 3 - e^x = 0$

2) $e^{2x-1} = e^{x^2}$

5) $1 + 5e^x - 6e^{2x} < 0$

3) $(3e^x + 1)(e^x - 2) = 0$

6) $e^{\ln^2 x - \ln x} = 1.$

Exercise 2 :

Determine the domain of definition and the derivative of each of the following functions:

1) $f(x) = 3e^x - 3x + 4$

6) $f(x) = \ln\left(\frac{e^x-1}{e^x+1}\right)$

2) $f(x) = \frac{2x+1}{e^x-1}$

7) $f(x) = (x-1)e^{0.5x} + x$

3) $f(x) = \ln(2e^x + 4) - x$

8) $f(x) = 2x + 3 - \frac{1}{-e^x-1}$

4) $f(x) = \frac{3}{e^{-x}+2}$

9) $f(x) = \frac{3e^x+4}{2e^x}$

5) $f(x) = \frac{e^x}{\sqrt{1-e^{2x}}}$

10) $f(x) = x^2e^{-x} + 3x \ln\left(\frac{1}{1-e^{-x}}\right).$

Exercise 3 :

Calculate the following limits:

1) $\lim_{x \rightarrow -\infty} \left(2 + e^{-x} + \frac{1}{x}\right)$

2) $\lim_{x \rightarrow 0^+} \left(\frac{3x-1}{e^x-1}\right)$

3) $\lim_{x \rightarrow +\infty} \left(\frac{3e^x-1}{e^x+1}\right)$

4) $\lim_{x \rightarrow +\infty} ((x-1)e^{-2x})$

5) $\lim_{x \rightarrow +\infty} \left(\frac{2x-2}{e^x-2x}\right)$

6) $\lim_{x \rightarrow +\infty} (e^{2x} - 2e^x)$

7) $\lim_{x \rightarrow 0^+} \left(\frac{e^{2x}-2e^x+3}{e^x-1}\right)$

8) $\lim_{x \rightarrow +\infty} (\ln(e^x + 3) - x)$

9) $\lim_{x \rightarrow -\infty} \left(-x + 2 - \frac{3e}{e^x+5}\right)$

10) $\lim_{x \rightarrow +\infty} (\ln(3e^x - \frac{1}{e^x}) - x)$

11) $\lim_{x \rightarrow +\infty} ((x^2-1)e^{-2x} - 3)$

12) $\lim_{x \rightarrow +\infty} \left(\frac{(x-1)e^x}{x}\right)$

13) $\lim_{x \rightarrow -\infty} ((x^2-2x+9)e^{-2x})$

14) $\lim_{x \rightarrow +\infty} (x^2 - 2e^x).$

Exercise 4 :

Calculate the following integrals:

1) $\int_0^1 (2x-3)e^{x^2-3x+4} dx$

6) $\int_0^1 (x-3)e^{2x} dx$

2) $\int_1^2 (x+2)e^{x^2+4x-1} dx$

7) $\int_0^1 (x^2e^x + 2) dx$

3) $\int_0^{\ln 2} \left(x - \frac{e^x}{e^x+2}\right) dx$

8) $\int_{-3}^{-1} \left(\frac{1+e^{-x}}{-x+e^{-x}}\right) dx$

4) $\int_0^1 (2e^{-4x} + 2) dx$

9) $\int_0^1 \frac{e^{2x}}{(3+e^{2x})^2} dx$

5) $\int_0^1 \left(\frac{1}{e^x+1}\right) dx$

10) $\int_{-3 \ln 2}^{-\ln 3} \left(\frac{e^{-x}}{\sqrt{e^{-x}+1}}\right) dx.$